

Macroeconomic Regime Identification & Tactical Asset Allocation

**Cornell Financial Engineering Manhattan
Financial Engineering Project (Final Presentation)**

**Sponsor: Vanguard Asset Management
Advisor: Prof. David Ruppert**

**Yichen Gao • Olivia Guo • Fanyi Lyu • Adarsh Pandey • Yixuan Wu • Yang Yue
Date: December 11, 2025**

Project Overview

Objective

Develop a systematic and quantitative approach for identifying macro-financial regimes and interpreting them through macro features and asset behavior.

As a secondary step, explore how regime information may enhance asset-allocation strategies.

Methods

1. Apply multiple unsupervised regime identification models to identify latent macro-financial states
2. Validate regime distinctiveness using macro-feature patterns and forward asset returns
3. Evaluate how these regime structures support and inform tactical asset allocation strategies

Key Findings

- Several models generate clear, macro-interpretable regime structures that align well with historical economic periods.
- Regimes show meaningful forward-return separation—particularly in equities, gold, and dollar assets—useful for informing asset allocation decisions
- Incorporating regime information into asset-allocation decisions enhances robustness of investor portfolios, offering improved downside protection and more consistent performance across market environments.

Data Overview

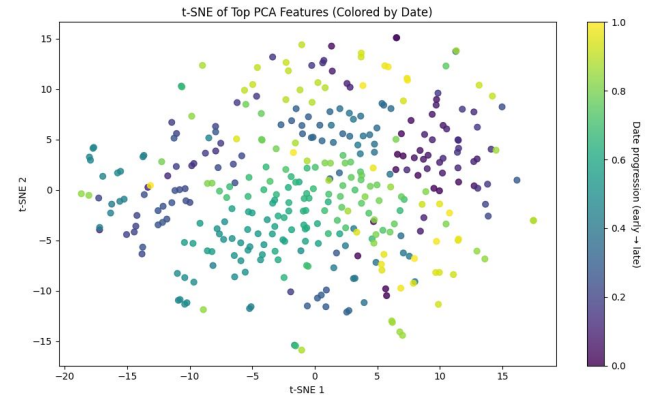
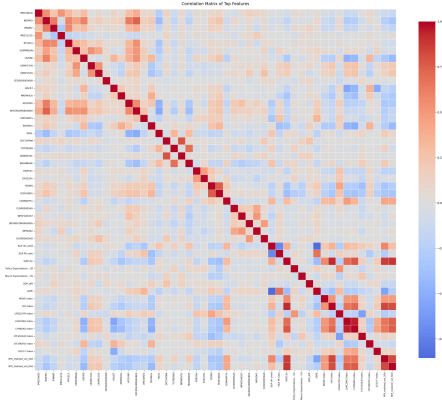
Dataset (1990-2025)

Group	Features	Count	Descriptions
① Output & Income	IPNCONGD, INDPRO, IPNMAT, IPB51222S, IPFUELS	5	IP Nondurable Consumer Goods • Industrial Production Index • IP Nondurable Materials • IP Residential Utilities • IP Fuels
② Labor Market	UEMPMEAN, USFIRE, UEMP27OV, UEMP15OV, CES0600000008	5	Avg Unemployment Duration • Employment (Financial Activities) • Unemployed 27+ Weeks • Unemployed 15+ Weeks • Avg Hourly Earnings (Goods-Producing)
③ Consumption & Orders	HOUST, AMDMUOX, DPCERA3M086SBEA, UMCSENTx, BUSINVx	5	Housing Starts • Unfilled Orders (Durables) • Real Personal Consumption Expenditures • Consumer Sentiment Index • Total Business Inventories
④ Money & Credit	M2SL, DTCTHFM, TOTRESNS, NONREVSL, BOGMBASE	5	M2 Money Stock • Total Consumer Loans and Leases • Total Reserves • Non-revolving Credit • Monetary Base
⑤ Interest Rate & FX Rates	EXJPUSx, EXSZUSx, TB3MS, FEDFUNDS, COMPAPFFx	5	Japan / US FX • Switzerland / US FX • 3-Month T-Bill Rate • Fed Funds Rate • Commercial Paper – Fed Funds Spread
⑥ Prices	CUSR0000SAS, WPSFD49207, DDURRG3M086SBEA, WPSID62, CUSR0000SAD	5	CPI Services • PPI Finished Consumer Goods • PCE Durables Deflator • PPI Durables • CPI Durables
⑦ Stock Market	S&P div yield, S&P PE ratio	2	S&P Composite Dividend Yield • S&P Price-Earnings Ratio
⑧ Other / Derived Indicators	GDP_diff, CAPE, MOVE Index, VIX Index, USGG2YR Index, LUACOAS Index, .0210 F Index, SPX_realized_vol_20d, SPX_realized_vol_60d	9	Derived or market indices – GDP gap, valuation (CAPE), bond/volatility measures, yield spreads, realized volatility metrics

Dataset (1990-2025)

- Used quantitative techniques like correlation pruning and PCA to boil down database from **140+ features** to **41 features**
- These are macro-economic and market features from **FRED_MD database** and some data from **Bloomberg**
- **32** features from **FRED**, **4** features from **Bloomberg**, **1 Shiller Cape** and **4 derived features**

- We see from the correlation heatmap that the newly pruned dataset has **fewer highly correlated variables**.
- The t-sne plot represents the data in a 2-D form and we can see that the clusters still **preserve temporal structure**, meaning that our dataset **retains its information**



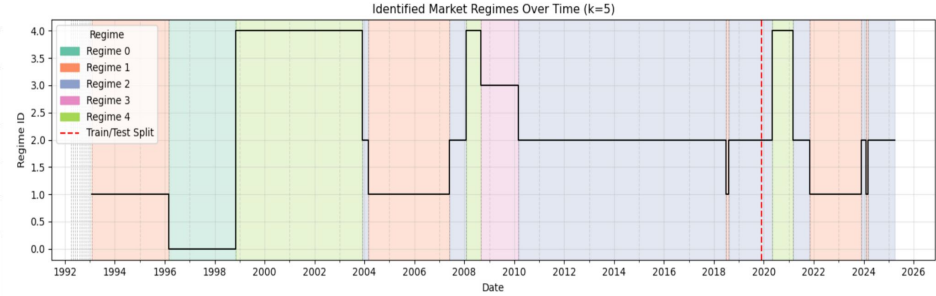
Regime Identification

Regime Identification Methods

	Time Series KMeans	Hidden Markov Model	Bayesian Gaussian Mixture Model	Regularized Jump Model
Algorithm Definition	Clusters rolling macro windows using K-Means with DTW distance. DTW aligns sequences to compare shape, not exact timing.	Probabilistic state-space model with hidden regimes; macro features generated by regime-specific distributions; transitions follow a first-order Markov process.	Probabilistic mixture model with soft regime membership; Gaussian components regularized by Dirichlet-process priors; tied covariance improves stability after whitening.	Centroid-based clustering enhanced with jump penalties to enforce temporal smoothness and feature-regularization penalties on regime centroids.
Key Characteristics	Captures structural macro patterns; Produces stable regimes.	Learns regime dynamics; captures evolution instead of static clustering; probabilistic regime decoding.	Captures overlapping regimes via soft assignments; handles uncertainty & smooth transitions; Bayesian shrinkage removes regimes and stabilizes estimates	Direct, interpretable feature selection via centroid shrinkage; stable regime paths through jump penalties; effective in high-dimensional settings.
Number regimes	K = 5 Selected using Elbow Method	K = 5 Selected using BIC	Max K = 5 Selected using BIC and validated through posterior shrinkage under DP prior	K = 4 Selected using Elbow Method
Calibration / Training	12-month rolling windows ; expanding-window standardization	EM estimation of regime parameters and transition matrix; feature-saliency weighting; apply smoothing / minimum-duration constraints	Variational Bayesian EM on whitened features; priors regularize means and weights; tied covariance improves robustness; probability smoothing and minimum-duration rules enforce persistence	Coordinate descent alternating between: 1. Cluster centroids 2. State Sequence

TS KMeans Regime Results

Regime ID	Label	Key Features
0	Mid-Cycle Stable Growth	Moderate growth; easy financial conditions; stable valuations; moderate volatility.
1	Healthy Expansion	Solid growth; low volatility; strong financial and liquidity backdrop.
2	Late-Cycle Softening	Slowing growth; neutral conditions; rising uncertainty.
3	Crisis / Shock	Negative growth; tight financial conditions; high volatility.
4	Early-Cycle Recovery	Improving growth; falling yields; easing volatility; strong liquidity support.



Regime	Name	GDP CYOY	CPI YOY	UNRATE	S&P 500 Return	PE	CAPE	Profit Margin	US Financial Conditions	VIX
0	Mid-Cycle Stable Growth	4.18	2.29	4.95	0.02	22.82	30.87	0.1	0.68	22.43
1	Healthy Expansion	3.19	3.74	5.07	0.01	21.19	25.72	0.11	0.93	15.22
2	Late-Cycle Softening	2.64	2.11	5.84	0.01	22.79	27.31	0.12	1.33	17.77
3	Crisis / Shock	-2.07	0.57	8.76	-0.01	41.58	17.27	0.09	3.57	34.49
4	Early-Cycle Recovery	2.05	2.46	5.4	0.0	29.5	32.21	0.09	1.6	24.73

Major turning points (2000, 2008, 2020) all map to the **same regime**, validating that the model reliably captures true macro stress periods.

Expansion, late-cycle softening, and recovery phases appear in the correct chronological order, showing realistic macro-cycle progression.

Out-of-sample regime assignments(2020) stay economically interpretable, highlighting the model's **robustness** to changing market conditions.

TS KMeans Regime Separability

Forward 6M Asset Returns by Market Regime (Train Sample, with ANOVA significance)

	Regime	Regime_Label	Russell 1000 (6M fwd)	US Agg Bond (6M fwd)	Gold (6M fwd)	Dollar (6M fwd)
0	0.000000	Mid-Cycle Stable Growth	0.128 ± 0.090	0.043 ± 0.020	-0.054 ± 0.043	0.032 ± 0.038
1	1.000000	Healthy Expansion	0.056 ± 0.057	0.024 ± 0.032	0.057 ± 0.090	0.006 ± 0.039
2	2.000000	Late-Cycle Softening	0.045 ± 0.079	0.020 ± 0.025	0.033 ± 0.114	0.010 ± 0.040
3	3.000000	Crisis / Shock	0.087 ± 0.166	0.045 ± 0.016	0.118 ± 0.066	-0.012 ± 0.054
4	4.000000	Early-Cycle Recovery	-0.030 ± 0.145	0.031 ± 0.029	0.028 ± 0.086	0.012 ± 0.049
5		ANOVA p-value	0.0000	0.0000	0.0000	0.0062

Forward 6M Asset Returns by Market Regime (Test Sample, with ANOVA significance)

	Regime	Regime_Label	Russell 1000 (6M fwd)	US Agg Bond (6M fwd)	Gold (6M fwd)	Dollar (6M fwd)
0	1.000000	Healthy Expansion	0.036 ± 0.116	-0.011 ± 0.052	0.049 ± 0.101	0.012 ± 0.039
1	2.000000	Late-Cycle Softening	0.076 ± 0.087	0.011 ± 0.035	0.133 ± 0.083	0.004 ± 0.040
2	4.000000	Early-Cycle Recovery	0.185 ± 0.051	-0.006 ± 0.016	-0.021 ± 0.069	-0.030 ± 0.029
3		ANOVA p-value	0.0005	0.1428	0.0000	0.0144

Forward 6M Relative Performance by Market Regime (Train Sample)

	Regime	Regime_Label	Growth - Value (6M fwd)	HY - IG (6M fwd)	Long - Short Treasury (6M fwd)	EM - DM Equities (6M fwd)
0	0.000000	Mid-Cycle Stable Growth	0.037 ± 0.06	-0.001 ± 0.041	0.027 ± 0.039	-0.098 ± 0.145
1	1.000000	Healthy Expansion	0.001 ± 0.045	0.017 ± 0.026	0.011 ± 0.051	0.051 ± 0.127
2	2.000000	Late-Cycle Softening	0.03 ± 0.053	0.005 ± 0.043	0.032 ± 0.078	-0.005 ± 0.061
3	3.000000	Crisis / Shock	0.026 ± 0.063	0.088 ± 0.1	0.01 ± 0.065	0.129 ± 0.089
4	4.000000	Early-Cycle Recovery	-0.024 ± 0.134	-0.02 ± 0.081	0.012 ± 0.046	0.038 ± 0.112
5		ANOVA p-value	0.0000	0.0000	0.0878	0.0000

Forward 6M Relative Performance by Market Regime (Test Sample)

	Regime	Regime_Label	Growth - Value (6M fwd)	HY - IG (6M fwd)	Long - Short Treasury (6M fwd)	EM - DM Equities (6M fwd)
0	1.000000	Healthy Expansion	0.033 ± 0.11	0.023 ± 0.02	-0.06 ± 0.08	-0.04 ± 0.039
1	2.000000	Late-Cycle Softening	0.078 ± 0.107	0.017 ± 0.022	-0.008 ± 0.059	-0.003 ± 0.061
2	4.000000	Early-Cycle Recovery	-0.033 ± 0.09	0.056 ± 0.026	-0.069 ± 0.063	0.003 ± 0.087
3		ANOVA p-value	0.0205	0.0000	0.0108	0.0380

Absolute Performance (6-month forward returns)

Train Sample (≤ 2019)

- Strong regime separation ($p \approx 0$).
- Equities strongest in stable growth; Gold & Bond lead in crisis.

Test Sample (> 2020)

- Regime separation effects persist but slightly weaker.
- Early-cycle recovery shows sharp equity rebound.

Relative Performance (6-month forward returns)

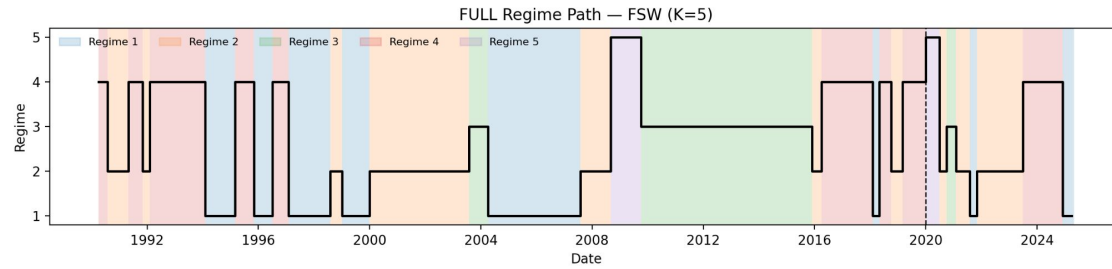
Train Sample (≤ 2019)

- Clear separation across spreads ($p \approx 0$).
- Growth > Value in most regimes; Value > Growth in early-cycle recovery.
- HY > IG in crisis; IG > HY in recovery.
- Long term bonds leads in most regimes.

Test Sample (> 2020)

- Separation remained clear ($p < 0.05$).
- Growth - Value pattern holds
- Short duration benefits during most regimes.

HMM Regime Results



Regime	Label	Historical Alignment / Interpretation	Key Characteristics	GDP CYOY	CPI YOY (Inflation)	UNRATE	Profit Margin	CAPE	S&P 500 Return	S&P PE ratio
1	Late Expansion / Soft Landing	1994–95 Fed soft landing, 2004–07 late-cycle expansion	Strongest GDP growth; Moderate inflation; Stable unemployment; High profit margins & firm valuations	3.7	2.74	4.92%	0.11	29.52	0.01	23.26
2	Late-Cycle Tightening	2000–03 post-dot-com tightening, 2021–23 inflation tightening cycle	Highest inflation; Slowing growth momentum; Flat labor conditions; Valuation compression	2.25	3.71	5.02%	0.1	29.1	0	25.35
3	Early Expansion / Risk-On Growth	2003–04 early-cycle rebound, 2020–21 post-COVID risk-on recovery	Low inflation; Improving labor market; Healthy but moderate GDP growth; Strong profit margins	2.1	1.69	7.48%	0.12	23.73	0.01	24.35
4	Mid-Cycle Normalization	1991–94 mid-cycle stabilization, 2019 pre-COVID normalization	Balanced inflation and growth; stable labor market; firm valuations; consistent equity performance	2.64	2.8	5.24%	0.1	25.85	0.01	22.8
5	Crisis/ Market Stress	2008–09 Global Financial Crisis, 2020 COVID shock	Contracting GDP; Sharp rise in unemployment; Very low inflation; Profit margin deterioration	-1.48	0.45	8.36%	0.09	19.98	-0.01	27.0

HMM Regime Separability

Objective: Test whether the regimes identified by the HMM correspond to statistically distinct market environments.

Method: ANOVA test on 6M forward returns

Result:

- Most assets show strong regime separation
 - regimes capture meaningful differences
- Equities / gold / dollar differentiate well
 - align with macro and risk sentiment conditions.
- Rate-spread & credit signals weaker (p-val > 0.05)
 - less relevant for policy-driven assets.
- Some sign flips in test set
 - OOS instability; regime effects weaken over time

Relative (6M fwd) — Train

Regime	R1000 Value - Growth	IG Corp - HY Corp	Short Tsy - Long Tsy	DM ex-US - EM	N
1	-0.032 ± 0.079	-0.006 ± 0.033	-0.015 ± 0.045	0.006 ± 0.139	94
2	0.026 ± 0.105	0.000 ± 0.094	-0.022 ± 0.051	-0.056 ± 0.118	82
3	-0.008 ± 0.036	-0.010 ± 0.041	-0.034 ± 0.081	0.018 ± 0.060	82
4	-0.023 ± 0.070	-0.005 ± 0.050	-0.020 ± 0.059	-0.039 ± 0.129	86
5	-0.041 ± 0.068	-0.115 ± 0.104	-0.012 ± 0.052	-0.148 ± 0.099	13
ANOVA p-value	0.0000	0.0000	0.0686	0.0000	

Absolute (6M fwd) — Train

Regime	Russell 1000	US Agg Bond	Gold	Dollar	N
1	0.067 ± 0.063	0.028 ± 0.029	0.034 ± 0.112	0.002 ± 0.038	94
2	-0.009 ± 0.155	0.040 ± 0.028	0.039 ± 0.101	0.013 ± 0.049	82
3	0.053 ± 0.081	0.019 ± 0.021	0.012 ± 0.104	0.015 ± 0.040	82
4	0.051 ± 0.073	0.031 ± 0.034	0.021 ± 0.073	0.024 ± 0.039	86
5	0.115 ± 0.183	0.046 ± 0.017	0.123 ± 0.072	-0.027 ± 0.054	13
ANOVA p-value	0.0000	0.0000	0.0031	0.0001	

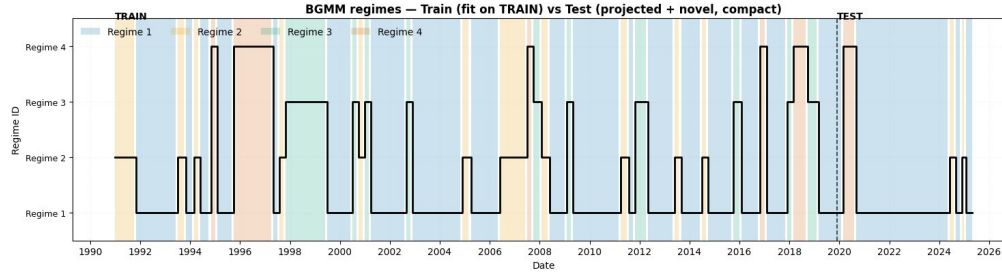
Absolute (6M fwd) — Test

Regime	Russell 1000	US Agg Bond	Gold	Dollar	N
1	0.009 ± 0.076	-0.017 ± 0.056	0.148 ± 0.080	-0.022 ± 0.050	6
2	0.037 ± 0.107	-0.019 ± 0.043	0.008 ± 0.086	0.010 ± 0.040	29
3	0.192 ± 0.067	-0.013 ± 0.010	-0.017 ± 0.067	-0.013 ± 0.022	4
4	0.099 ± 0.074	0.027 ± 0.021	0.156 ± 0.053	0.017 ± 0.026	17
5	0.186 ± 0.102	0.028 ± 0.017	0.137 ± 0.087	-0.040 ± 0.037	6
ANOVA p-value	0.0005	0.0008	0.0000	0.0081	

Relative (6M fwd) — Test

Regime	R1000 Value - Growth	IG Corp - HY Corp	Short Tsy - Long Tsy	DM ex-US - EM	N
1	0.014 ± 0.086	-0.023 ± 0.020	0.045 ± 0.047	0.018 ± 0.037	6
2	-0.024 ± 0.118	-0.027 ± 0.026	0.058 ± 0.088	0.038 ± 0.054	29
3	0.065 ± 0.053	-0.056 ± 0.026	0.079 ± 0.051	0.056 ± 0.062	4
4	-0.056 ± 0.040	-0.019 ± 0.010	0.014 ± 0.041	0.011 ± 0.045	17
5	-0.194 ± 0.117	-0.027 ± 0.045	-0.015 ± 0.071	-0.090 ± 0.027	6
ANOVA p-value	0.0005	0.1417	0.0731	0.0000	

BGMM Regime Results



Train from 1990-01-31 to 2019-12-01

Regime	Category	n_months	pct_of_total
1	Mid Expansion	205	58.91%
2	Steady State	54	15.52%
3	Steady State	54	15.52%
4	Crisis	35	10.06%

Test from post 2019-12-01

Regime	Category	n_months	pct_of_total
1	Mid Expansion	54	83.08%
2	Steady State	5	7.69%
4	Crisis	6	9.23%

- Crisis
- Steady State
- Steady State
- Mid Expansion

Regime	S&P 500 Return	US Financial Conditions	VIX	Profit Margin	CPI YoY	GDP CYoY	CAPE	S&P PE Ratio	Unemployment Rate
0	0.01	1.34	19.20	0.11	2.66	2.52	26.86	25.70	5.79
1	0.01	1.14	16.97	0.10	3.20	2.11	25.07	20.22	5.74
2	0.01	1.61	24.07	0.10	2.06	2.99	30.88	24.66	5.15
3	0.01	0.97	18.82	0.10	2.43	2.32	27.13	21.08	5.76

- 1. Crisis Detection** – Accurately flags COVID-19 as a crisis using only pre-2020 training data, showing strong generalization to unseen shocks.
- 2. Post-COVID Shift** – Mid-Expansion dominates the test period, reflecting moderate returns with elevated volatility but insufficient calm for Steady State.
- 3. Stable Regimes** – Regime durations span 3–12 months with no rapid switching, confirming smooth and economically sensible transitions.
- 4. Out-of-Sample Robustness** – Test regimes remain interpretable even with frozen parameters, indicating the model captures structural, not noise-driven, patterns.

Economic Regime:

- **Crisis:** Severe downturn marked by large negative returns and sharp spikes in volatility or market stress
- **Correction:** Moderate pullback with rising volatility but without systemic stress
- **Walking on Ice:** Very strong returns occurring alongside elevated volatility or stress
- **Steady State:** Healthy growth environment with positive returns, low volatility, and low stress—ideal conditions for stable risk-taking
- **Inflation:** Positive but uncertain environment where returns remain strong but volatility or financial stress is elevated
- **Mid Expansion:** Neutral baseline regime with moderate returns and muted volatility and stress



BGMM Regime Separability

Train from 1990-01-31 to 2019-12-01

Forward 12M Relative Performance by Market Regime (Train Sample)

Regime	Regime_Label	Growth – Value (12M fwd)	HY – IG (12M fwd)	Long – Short Treasury (12M fwd)	EM – DM Equities (12M fwd)
0	Mid Expansion	-0.006 ± 0.11	0.028 ± 0.078	0.042 ± 0.082	0.034 ± 0.145
1	Steady State	0.04 ± 0.089	0.011 ± 0.096	0.042 ± 0.075	0.128 ± 0.237
2	Steady State	0.049 ± 0.174	0.007 ± 0.117	0.011 ± 0.074	0.045 ± 0.216
3	Crisis	0.047 ± 0.052	0.019 ± 0.039	0.05 ± 0.064	-0.049 ± 0.123
ANOVA p-value		0.0013	0.3142	0.0526	0.0000

Forward 12M Asset Returns by Market Regime (Train Sample, with ANOVA significance)

Regime	Regime_Label	Russell 1000	US Agg Bond	Gold	Dollar
0	Mid Expansion	0.091 ± 0.137	0.056 ± 0.043	0.088 ± 0.16	0.019 ± 0.06
1	Steady State	0.025 ± 0.163	0.067 ± 0.046	0.064 ± 0.156	0.03 ± 0.077
2	Steady State	0.095 ± 0.197	0.05 ± 0.039	0.039 ± 0.107	0.012 ± 0.045
3	Crisis	0.193 ± 0.152	0.077 ± 0.04	-0.015 ± 0.159	0.046 ± 0.057
ANOVA p-value		0.0000	0.0084	0.0015	0.0377

Test from post 2019-12-01

Forward 12M Relative Performance by Market Regime (Test Sample)

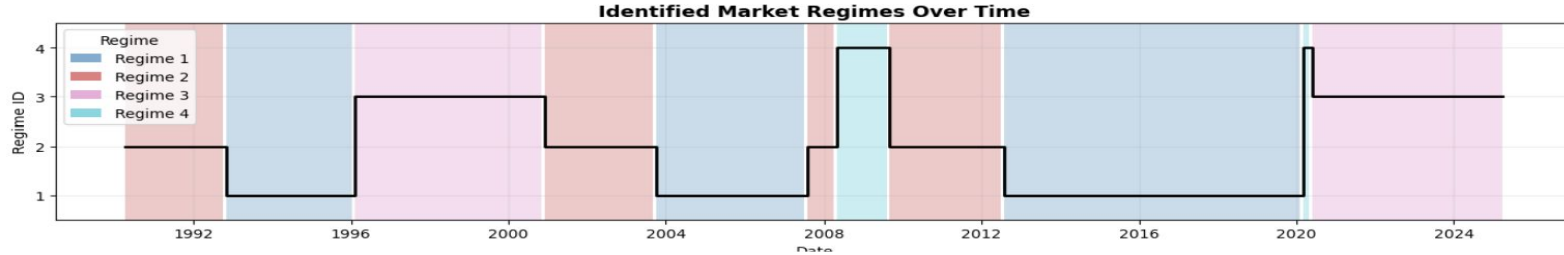
Regime	Regime_Label	Growth – Value (12M fwd)	HY – IG (12M fwd)	Long – Short Treasury (12M fwd)	EM – DM Equities (12M fwd)
0	Mid Expansion	0.073 ± 0.151	0.047 ± 0.023	-0.078 ± 0.083	-0.059 ± 0.072
1	Steady State	nan ± nan	nan ± nan	nan ± nan	nan ± nan
3	Crisis	0.063 ± 0.097	0.116 ± 0.033	-0.123 ± 0.032	0.072 ± 0.087
ANOVA p-value		0.8829	0.0000	0.1899	0.0001

Forward 12M Asset Returns by Market Regime (Test Sample, with ANOVA significance)

Regime	Regime_Label	Russell 1000	US Agg Bond	Gold	Dollar
0	Mid Expansion	0.121 ± 0.164	-0.011 ± 0.067	0.126 ± 0.151	0.025 ± 0.041
1	Steady State	0.138 ± 0.014	0.05 ± 0.014	0.389 ± 0.026	-0.016 ± 0.019
3	Crisis	0.424 ± 0.092	0.001 ± 0.008	0.033 ± 0.062	-0.068 ± 0.022
ANOVA p-value		0.0002	0.2597	0.0032	0.0000

- 1. Crisis Recovery Premium** – Crisis regimes deliver the strongest forward equity returns, validating contrarian positioning and reflecting powerful policy-driven recoveries.
- 2. Defensive Asset Shifts** – Bonds lose regime differentiation post-2020 as rates become policy-driven, while gold switches from crisis underperformance to safe-haven gains.
- 3. Factor Rotation Reversal** – Growth leads in past crises, but Value leads in recent ones, reflecting the shift from low-rate secular growth to inflation-driven cyclical leadership.
- 4. Persistent Credit Risk** – HY-IG spreads remain consistently significant across periods, confirming credit risk as a stable, regime-sensitive return driver.

Jump Model Regime Results



Economic Meanings and Key Features (full sample)

Regime	Label	Key Traits (Short Version)	Profit Margin	CAPE Valuation	S&P500 Return	CPI YoY	GDP CYoY	Unemployment	S&P PE Ratio	VIX Index	LUACOAS Index
1	Low-Inflation Stable Expansion	Stable growth, low inflation, steady equity gains.	0.11	25.63	0.01	2.21	2.85	5.39	21.42	14.54	1.06
2	Inflationary Tightening / Late-Cycle	Inflation rising, growth softening, tightening pressure.	0.09	22.5	0.00	2.95	1.58	6.91	25.26	22.42	1.58
3	Overheating / Pre-Recession Slowdown	High inflation, stretched valuations, volatile growth.	0.11	34.22	0.01	3.40	3.40	4.62	27.01	21.74	1.03
4	Crisis / Recessionary Stress	GDP contraction, rising unemployment, falling equities.	0.09	18.88	-0.02	1.30	-0.98	8.09	23.7	35.51	3.7

Jump Model Regime Separability

Absolute performance					Relative Performance				
Regime	Russell 1000	US Agg Bond	Gold	Dollar	Regime	Growth – Value	HY – IG	Long – Short Treasury	EM – DM Equities
1-train	0.110 ± 0.093	0.043 ± 0.043	0.074 ± 0.159	0.021 ± 0.059	1-train	0.001 ± 0.017	-0.002 ± 0.012	0.003 ± 0.023	-0.001 ± 0.017
1-test	0.090 ± 0.095	0.045 ± 0.082	-0.010 ± 0.145	0.028 ± 0.060	1-test	0.001 ± 0.018	-0.002 ± 0.013	0.004 ± 0.024	-0.002 ± 0.018
2-train	0.043 ± 0.179	0.080 ± 0.039	0.098 ± 0.140	0.011 ± 0.060	2-train	-0.000 ± 0.030	-0.006 ± 0.027	0.004 ± 0.026	-0.012 ± 0.036
2-test	0.145 ± 0.178	0.072 ± 0.136	-0.006 ± 0.176	0.035 ± 0.062	2-test	-0.001 ± 0.031	-0.007 ± 0.029	0.005 ± 0.027	-0.010 ± 0.035
3-train	0.146 ± 0.174	0.034 ± 0.068	0.025 ± 0.157	0.035 ± 0.051	3-train	0.005 ± 0.041	-0.002 ± 0.014	-0.003 ± 0.028	-0.005 ± 0.020
4-train	0.093 ± 0.319	0.071 ± 0.034	0.173 ± 0.138	0.001 ± 0.083	4-train	0.014 ± 0.031	-0.004 ± 0.050	0.008 ± 0.044	0.015 ± 0.039
4-test	0.032 ± 0.310	0.010 ± 0.122	-0.014 ± 0.215	0.006 ± 0.085	4-test	0.013 ± 0.030	-0.003 ± 0.052	0.007 ± 0.046	0.014 ± 0.040
ANOVA p-value (train)	0.0000	0.0000	0.0001	0.0089	ANOVA p-value (train)	0.1302	0.6643	0.0713	0.4082
ANOVA p-value (test)	0.019	0.034	0.028	0.041	ANOVA p-value (test)	0.1423	0.5738	0.0834	0.4687

Result Comparison

	TS KMeans	HMM	BGMM	Regularized Jump Model
Pros	Macro-consistent regimes with strong out-of-sample crisis detection; Meaningful forward-return separation.	Clear macro-consistent regimes; strong alignment with historical cycles and broad asset behavior; good forward-return separability for most asset classes.	Soft regime probabilities; Automatic complexity control; Stable estimation; Robust to small regimes; Interpretable uncertainty	Robust regime classification with smooth state sequences and strong interpretability from feature-selection.
Cons	Sensitive to window choices; no probabilistic state uncertainty.	Sensitive to model set up → not robust; train–test forward-return patterns diverge → effectiveness decay over time.	Higher computational cost; Sensitive to preprocessing; Less sharp segmentation; Hyperparameter dependent	Computationally expensive and shows weaker relative-performance results.
Forward looking return implications	Overall good separability, offers useful insights for regime-aware allocation; Less effective in distinguishing rate-spread.	Regimes offer differentiated forward-return patterns useful for regime-aware allocation; strongest signals in equities/gold/dollar, weaker and less stable in rate-spread and credit assets.	It distinguishes equity, gold, and dollar well, but provides weaker separation for rate-sensitive assets; useful for stable, uncertainty-aware regime-allocation but less sharp for cross-asset relative-value signals.	Regimes effectively separate absolute asset-class returns but perform less well in distinguishing relative returns.

Regime-Aware Asset Allocation

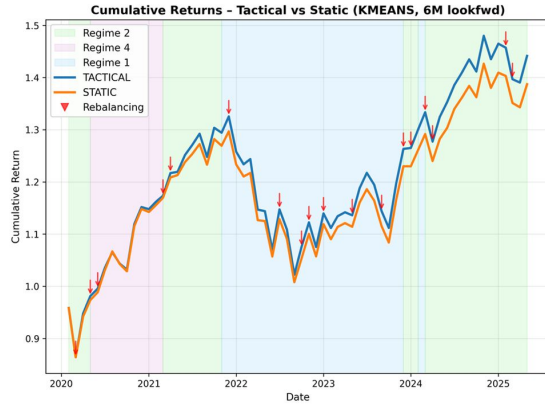
Asset Allocation Summary

	Rel. Performance Based AA	Regime-Based Mean-Var Optimization (MVO)	Regime Aware Benchmark Tilts(RABT)	Regime-Aware Black Litterman
Definition	Uses 6-month forward relative risk premiums to generate signals to allocate tactical weights to asset classes, while enforcing constraints. More Premium=More Weight	Uses regime-conditioned expected returns and covariances to find weights that maximize return-risk under investment constraints	Tilt benchmark weights up/down based on regime-specific forward return for each asset class, while enforcing tight constraints around the benchmark.	Combines regime-conditional equilibrium priors with structured investor views via Bayesian updating. Blends parameters across regimes using posterior probabilities.
Pros	Using relative performance rather than actual expected returns leads to more intuitive tactical allocation .	- Fully systematic; - Learns from history; - Interpretable	Highly stable and pragmatic ; easy to implement. Limits turnover and deviation from benchmark.	- Balances data-driven equilibrium with judgment - Accounts for regime uncertainty - Reduces estimation error
Cons	Less systematic than MVO or Black-Litterman, does not take risk into consideration.	- Sensitive to estimation error; - Purely data-driven; - lacks financial/economic judgment overlay	Tilts are small , may not fully capture strong regime opportunities; Relies on quality of regime signals, limited upside if signals are weak.	- Requires careful specification of views (P, q) and confidence (Ω) per regime - Computationally more complex

Benchmark

- We use a classic 60/40 portfolio as our benchmark.
- **60/40 Portfolio Breakdown:**
- **Equities (60%)**
 - 25% Russell 1000 Value – stable, dividend-paying large caps
 - 25% Russell 1000 Growth – innovative, high-growth firms
 - 10% Russell 2000 – small cap
- **Bonds (40%)**
 - 10% Short-term Treasuries – liquidity and safety
 - 10% Long-term Treasuries – duration and diversification
 - 15% IG Corporate Bonds – moderate yield, strong credit
 - 5% HY Corporate Bonds – small allocation for extra return
- **Constraints**
 - no shorting
 - **Asset class level:** Keep deviations within **10%**.
 - **Sub-asset class level** (e.g., value vs. growth, large cap vs. small cap): Keep deviations within **5%**.

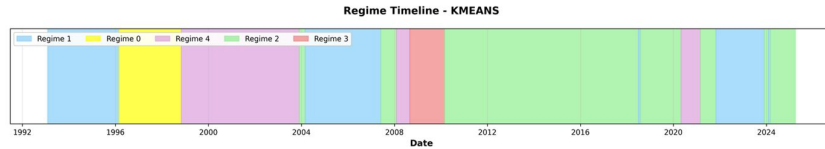
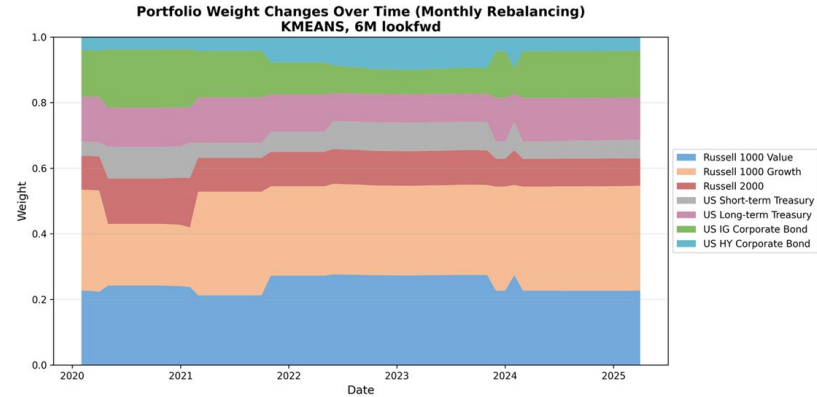
Relative Perf based K-means performance



Excess Performance Summary

Metric	Value
Excess Return	0.78%
Tracking Error	0.94%
Information Ratio	0.83

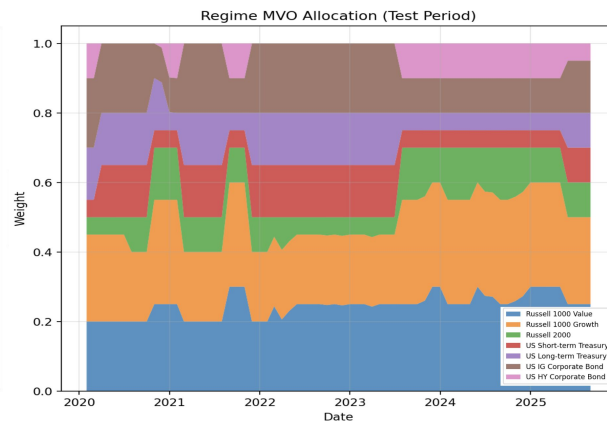
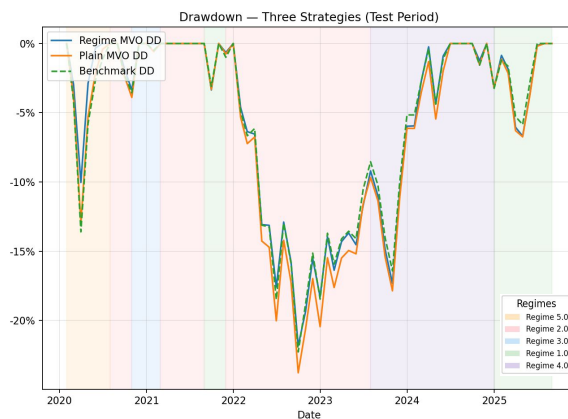
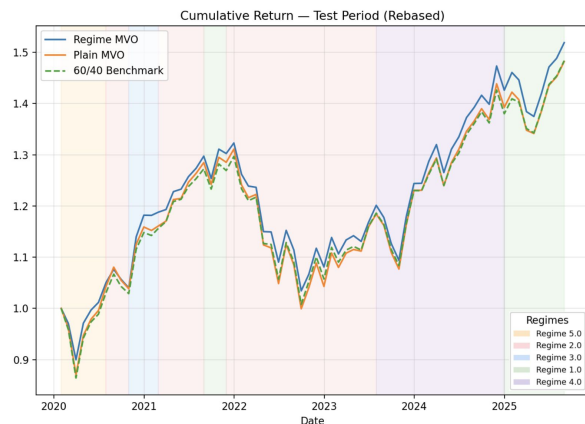
Metric	TACTICAL	STATIC
Ann. Return	7.83%	7.05%
Ann. Vol	13.93%	13.48%
Sharpe	0.562	0.523
Max DD	22.87%	22.28%



- Does well because of correct and stable allocation, regime 2 (green) is stable expansion, regime 1 (blue) is slow correction, regime 4 (purple) is crisis

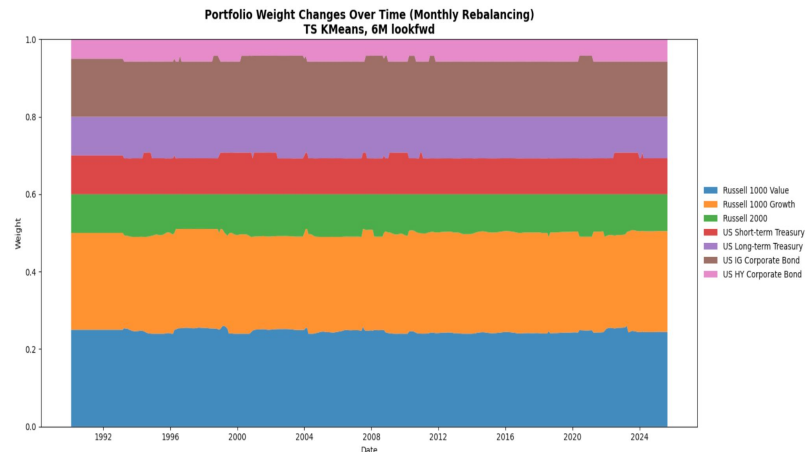
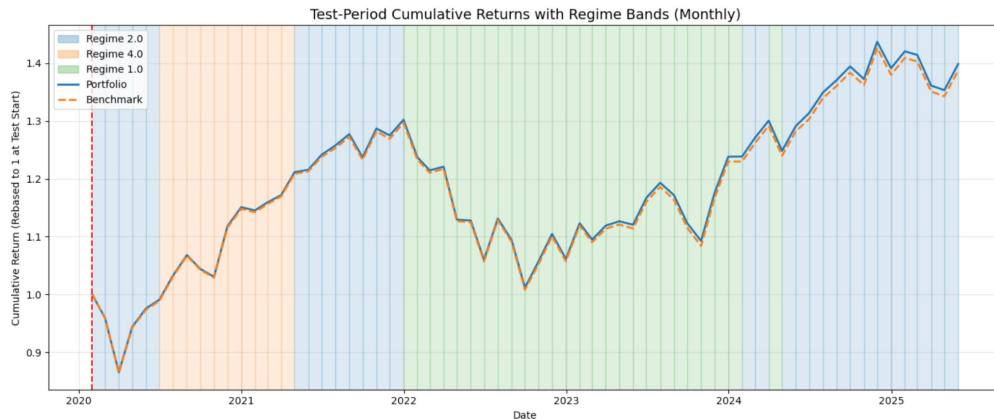
Regime-Based MVO Results (HMM)

In both train and test set, regime-based MVO beat both plain MVO and benchmark.
But the performance weaken in test set.



	Full (Regime MVO)	Full (Plain MVO)	Full (Benchmark)	Train (Regime MVO)	Train (Plain MVO)	Train (Benchmark)	Test (Regime MVO)	Test (Plain MVO)	Test (Benchmark)
Annual Return	0.091478815	0.083827517	0.080625814	0.092849592	0.084397861	0.080798833	0.084241922	0.080816439	0.079712379
Annual Volatility	0.096486308	0.102813198	0.099446133	0.090307423	0.09555284	0.092405135	0.124087843	0.134807672	0.130464586
Sharpe (rf=0)	0.948101516	0.815338099	0.810748615	1.028150168	0.883258528	0.874397647	0.6788894	0.599494361	0.610988632
Max Drawdown	-0.301093028	-0.353567952	-0.352765824	-0.301093028	-0.353567952	-0.352765824	-0.218261721	-0.237834415	-0.222792269
Tracking Error	0.016889903	0.007023804	-	0.015969886	0.006438565	-	0.020997867	0.00951276	-
Information Ratio	0.642573298	0.45583604	-	0.754592674	0.558979849	-	0.215714409	0.116060952	-
Excess Return	0.010853001	0.003201703	-	0.012050759	0.003599028	-	0.004529542	0.00110406	-

Regime Aware Benchmark Tilt Result



	Train Portfolio			Test Portfolio			Full Portfolio		
	Train Portfolio	Train Benchmark	Train Excess	Test Portfolio	Test Benchmark	Test Excess	Full Portfolio	Full Benchmark	Full Excess
Annualized Return	7.79%	7.62%	—	7.25%	7.08%	—	7.70%	7.53%	—
Volatility (Ann)	9.20%	9.20%	—	13.36%	13.37%	—	10.00%	10.00%	—
Sharpe Ratio	0.85	0.83	—	0.54	0.53	—	0.77	0.75	—
Max Drawdown	34.96%	35.28%	—	22.33%	22.28%	—	34.96%	35.28%	—
Excess Return (Ann)	—	—	0.17%	—	—	0.17%	—	—	0.17%
Tracking Error (Ann)	—	—	0.13%	—	—	0.17%	—	—	0.13%
Information Ratio	—	—	1.38	—	—	0.96	—	—	1.27

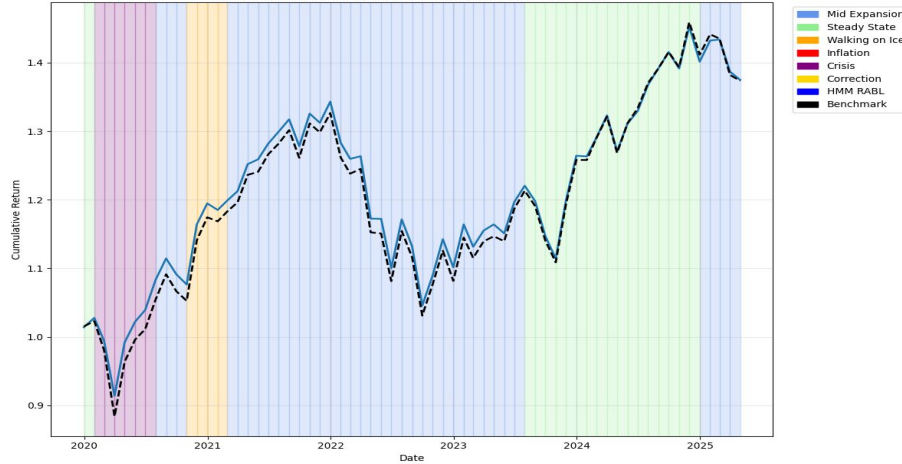
Consistently **outperforms the benchmark** with slightly lower volatility and a higher Sharpe ratio.

Delivers a strong **Information Ratio of 0.96**, indicating efficient risk-adjusted alpha.

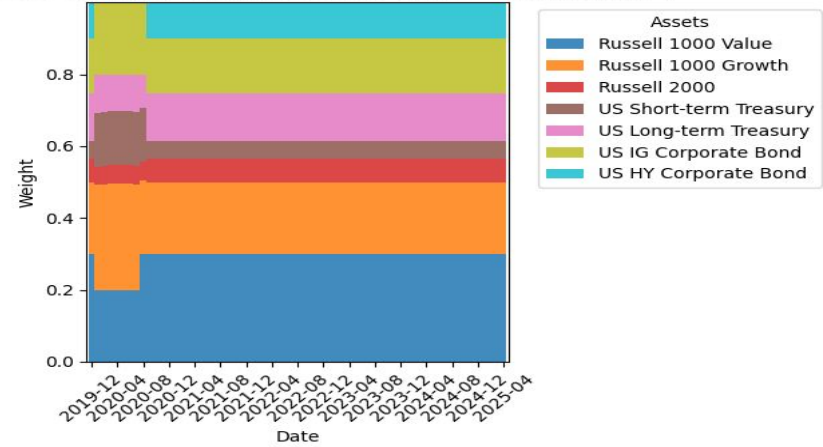
Stable across train/test periods, suggesting robust generalization.

Regime Aware Black Litterman Result

HMM RABL vs Benchmark with Regime Overlay



Portfolio Weight Changes Over Time — HMM RABL



Key Findings:

- Soft regime assignments improve robustness
- HMM dominates in risk-adjusted space

The smoother transition structure of BGMM and HMM provides more coherent risk-taking, while K-Means' discontinuous regime labels induce unnecessary de-risking during regime noise.

CornellEngineering

Operations Research and Information Engineering
Financial Engineering Manhattan

Metric	BGMM	HMM	KMeans	Benchmark
Annualized Return	6.42%	6.54%	5.42%	6.57%
Annualized Volatility	12.91%	12.85%	12.58%	13.42%
Sharpe Ratio	0.497	0.509	0.431	0.490
Max Drawdown	-22.22%	-22.16%	-22.03%	-22.28%
Annualized Excess Return	-0.16%	-0.04%	-1.15%	—
Tracking Error (Ann.)	1.12%	1.22%	1.47%	—
Information Ratio	-0.140	-0.030	-0.783	—
Cumulative Return	36.44%	37.49%	29.69%	37.36%



Conclusion & Recommendations

Conclusion

Regime-aware asset allocation builds portfolios that adjust risk in line with underlying macro conditions.

By becoming defensive in crisis regimes and more risk-seeking in expansions, the portfolio improves downside protection while capturing more upside when conditions allow.

Across regime models and allocation approaches, this dynamic posture leads to more resilient, more adaptive, and better risk-adjusted performance than traditional static strategies.

Recommendation

Overall, we recommend four regime–allocation pairs as the most practical and effective combinations for real-world implementation:

- TS KMeans + Relative Performance Based Approach
- TS KMeans + Regime-Aware Benchmark-Tilt Strategy
- HMM + Regime-Based MVO
- HMM + Regime-Aware Black Litterman Strategy (different investors' views input can affect performance)

These pairings consistently demonstrate strong performance, robustness across market conditions, and clear economic intuition, making them high-value candidates for further exploration and deployment.

Limitation & Next Step

Limitation

- Regime boundaries learned only up to 2019 degrade in 2020–2025
- Static regime definitions lose relevance as new data arrives
- Fixed number of regimes becomes inadequate as market shocks introduce new structure
- Stale regime labels
 - weaker regime-conditional forward returns
 - declining tilt effectiveness

Next Step

Adopt expanding-window regime identification:

Retrain regimes periodically; update boundaries with new data; use centroid matching for stable labels

Deploy fully adaptive RABT:

Combine expanding TS-KMeans with benchmark-tilt allocation; improve persistence and real-time robustness.

Allow dynamic regime complexity:

Allow time-varying number of regimes; select via model-selection or stability diagnostics to capture structural breaks.

Enhance RABL:

data-driven view generation with statistical confidence intervals and factor-based view extensions

Expanding TS KMeans + RABT

Cumulative Returns (Monthly)
Expanding TS-KMeans Regime-Aware Benchmark-Tilt Strategy



	Portfolio	Benchmark	Excess
Annualized Return	7.27%	7.09%	—
Volatility (Ann)	10.35%	10.34%	—
Sharpe Ratio	0.70	0.69	—
Max Drawdown	34.84%	35.28%	—
Excess Return (Ann)	—	—	0.18%
Tracking Error (Ann)	—	—	0.13%
Information Ratio	—	—	1.40

Overall Performance (From 2005 to 2025)

- Delivers higher returns than the benchmark
- Volatility is similar to the benchmark
- Smaller max DD to the benchmark
- Produces a higher Sharpe ratio
- Achieves a strong Information Ratio of 1.4, indicating efficient risk-adjusted excess return

THANK YOU

Technical Appendix

Relative Performance Based Asset Allocation

Step 1: Measure relative performance Based on 5 Market Premia with 6-month forward looking premium

1. Equity Risk Premium (Equities vs Short-term Treasuries)
2. Value Premium (Value vs Growth)
3. Credit Premium (High Yield vs Investment Grade)
4. Term Premium (Long-term vs Short-term Treasuries)
5. Size Premium (Small-Cap vs Large-Cap)

Step 2: Weight Different Tilts (over/underallocate)

1. Total tilt budget: 20% of portfolio value available for tactical adjustments
2. Use softmax normalization to allocate budget based on signal strength
3. Budget Allocation Example:
 - a. Equity tilt: $20\% \times 50\% = 10\%$ (strongest signal)
 - b. Value tilt: $20\% \times 25\% = 5\%$ (second strongest)
 - c. Credit tilt: $20\% \times 8\% = 1.6\%$
 - d. Term tilt: $20\% \times 8\% = 1.6\%$
 - e. Size tilt: $20\% \times 9\% = 1.8\%$

Step 3: Constrained Optimization

1. Raw tilts may violate investment constraints
2. Optimization objective: Minimize deviation from target while respecting rules
3. Hard Constraints:
 - a. No short positions (all weights $\geq 0\%$)
 - b. Weights sum to 100%
 - c. Total equity between 50-70%
4. Soft Penalties:
 - a. Minimize distance from signal-based targets
 - b. Penalize excessive deviation from strategic base

KMeans: Pre-2020: 6-Month Forward Returns by Regime

	Sizeprem	creditprem	equityrp	termprem	val-grwth
0.0	-7.72%	-0.10%	9.11%	2.67%	-3.57%
1.0	0.45%	1.63%	3.52%	1.06%	-0.13%
2.0	-0.78%	0.49%	3.37%	3.11%	-2.92%
3.0	2.61%	8.00%	7.24%	1.02%	-2.45%
4.0	4.31%	-2.01%	-5.61%	1.14%	2.81%

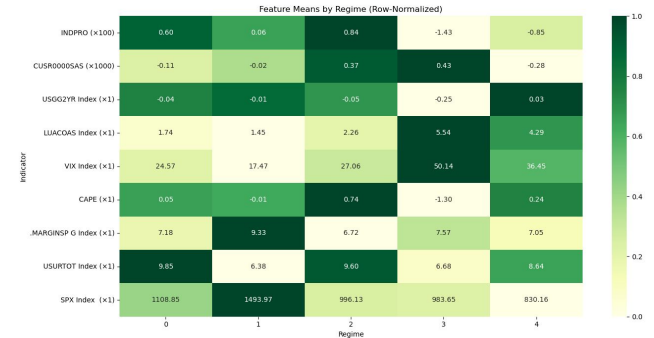
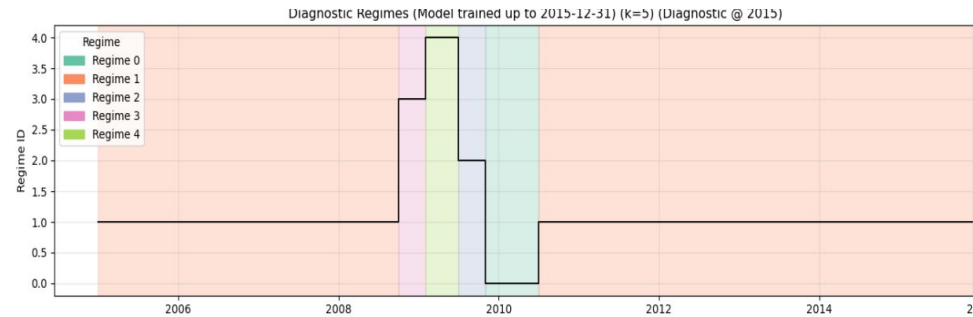
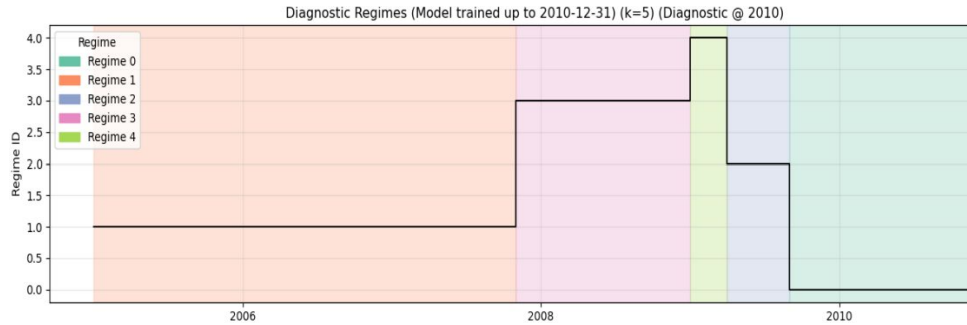
KMeans: 2020+: 6-Month Forward Returns by Regime

	Sizeprem	creditprem	equityrp	termprem	val-grwth
1.0	-2.72%	2.26%	3.35%	-6.01%	-2.54%
2.0	-5.98%	1.71%	6.12%	-0.65%	-6.69%
4.0	9.62%	5.57%	18.68%	-6.86%	2.70%

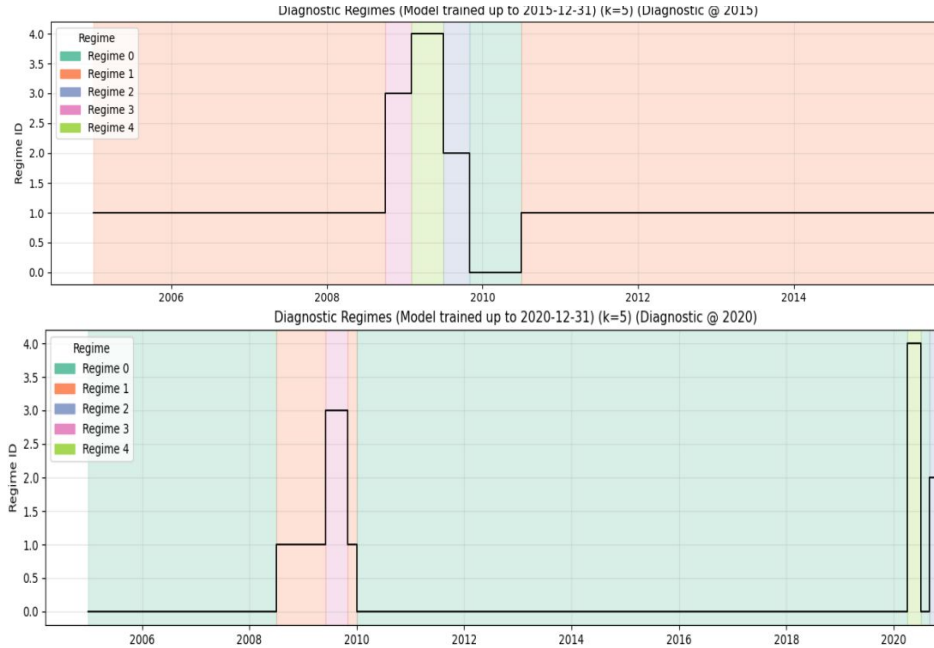
Regime Aware Benchmark Tilt Methodology

- Tradable Regime Signal
 - Shift regime labels forward by one month to form a “Regime Signal” that can be known at portfolio decision time.
 - Use this lagged signal for allocation decisions to prevent lookahead bias.
- Forward-Return Profiling
 - Compute 6-month forward returns for each asset class.
 - For each regime, calculate historical means and sample counts of forward returns.
 - Apply a shrinkage adjustment toward the global mean to stabilize noisy or under-sampled regimes.
- Constraints:
 - No shorting
 - $\pm 5\%$ deviation per asset
 - $\pm 10\%$ deviation per group (Equity, Rates, Credit)
- Signal-to-Weight Mapping
 - Within each group, convert regime-conditional forward returns into standardized z-scores.
 - Map signals to target tilts ($\pm 5\%$) and solve a quadratic optimization minimizing distance to target while satisfying constraints.
- Rebalancing
 - Monthly rebalancing on the first business day

Stability of TS K-Means with Expanding Window



Stability of TS K-Means with Expanding Window



Regime Mapping Between 2020 and 2015

2020 Regime	Matching 2015 Regimes
0	1, 0
1	3, 4, 0
2	None
3	2
4	None

Comparison of 2015 Regime 2 vs. 2020 Regime 3

Feature	2015 – Regime 2	2020 – Regime 3
Industrial Production	0.84 ± 0.43	0.61 ± 0.64
Core CPI (MoM) $\times 1000$	0.37 ± 1.31	0.46 ± 1.15
2Y Treasury Yield	-0.05 ± 0.06	-0.01 ± 0.12
US Financial	2.26 ± 0.18	2.42 ± 0.39
VIX Volatility $\times 1$	27.06 ± 2.43	26.92 ± 2.13
CAPE Valuation $\times 1$	0.74 ± 0.47	0.67 ± 0.44
Margin Debt Growth $\times 1$	6.72 ± 0.07	6.73 ± 0.09
Unemployment Rate $\times 1$	9.6 ± 0.14	6.68 ± 0.51
SPX Index Level $\times 1$	996.13 ± 58.57	983.65 ± 126.11